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AI-Powered Real Estate Intelligence: A Portfolio Dashboard for Smarter Decisions

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Project Domain

[Excel / Power BI with Copilot]

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

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3. Problem Statement

The Indian real estate sector manages large and complex property portfolios across multiple cities, yet most sales and portfolio decisions continue to rely on disconnected spreadsheets, manual reports, and isolated data exports rather than a single consolidated view. This creates a critical visibility gap — managers cannot identify stalled assets, track pricing effectiveness by location, or monitor agent performance in real time. Based on a real-world dataset of 550 property records across 8 Indian cities — Ahmedabad, Bangalore, Chennai, Delhi, Hyderabad, Kolkata, Mumbai, and Pune — spanning February 2024 to February 2026, five critical operational challenges were identified:

1. No Centralized Dashboard to Monitor Sales Performance Across 8 Cities: Sales managers have no single unified view to compare city-wise revenue, conversion rates, or agent output simultaneously. Bangalore generates ₹111.9 Cr while Delhi trails at ₹68.1 Cr — a 64% revenue gap that remains invisible without a centralized location-level dashboard.

2. Pricing Strategy Misaligned with City-Level Buyer Demand: Properties are listed without benchmarking against city-specific demand patterns. Hyderabad holds the highest average sale price at ₹1.51 Cr/unit yet records the lowest sold rate of just 40.8% among all 8 cities, while Kolkata (55.9%) and Chennai (55.7%) achieve far better conversions at lower price points.

3. Zero-Income Vacant Inventory Undetected Without an Early-Warning Flag: Of 550 properties, 281 (51.1%) remain unsold with locked inventory value of ₹416.3 Cr — exceeding realized sales revenue of ₹361.7 Cr. Within this, 136 properties are both Vacant and Unsold, representing ₹206.1 Cr in zero-income assets with no priority flag or corrective action mechanism.

4. Agent Productivity Imbalance with No Performance Accountability Mechanism: The top 5 agents generated ₹230.9 Cr — 64% of total portfolio revenue — while bottom-quartile agents averaged under ₹10 Cr each. The full commission pool of ₹7.46 Cr is disbursed with no ranking or performance-visibility layer, making it impossible to reward top performers or redirect underperformers.

5. Seasonal Buying Cycles Invisible Without a Monthly Trend Dashboard: Monthly data shows consistent demand peaks — 17 transactions (₹27.6 Cr) in May 2025 and 17 transactions (₹21.5 Cr) in October 2024, aligned with India's Navratri–Diwali cycle. November through January drops to just 5–6 sales per month — a 65% volume decline — invisible without a monthly trend view, causing campaigns and launches to be timed arbitrarily.

This project addresses all five challenges by building a Real Estate Sales and Property Portfolio Dashboard using Microsoft Excel enhanced with Microsoft Copilot. The dashboard consolidates 550 property records, tracks key performance indicators, and generates AI-

assisted insights to help stakeholders make faster, more informed business decisions. By combining structured data analysis with AI integration, the project demonstrates how modern tools can transform raw real estate data into a powerful decision-support system.

4. Objectives

The following objectives define the specific goals of the Real Estate Sales and Property Portfolio Dashboard project.

- To explore and understand the real estate dataset comprising 550 property records, including fields such as Property ID, Location, Property Type, Sale Price, Agent Name, Commission %, and Occupancy Status.
- To clean and preprocess the dataset by standardizing price and date formats, removing duplicate property records, handling missing values, and normalizing categorical fields such as Property Type and Occupancy Status.
- To compute key performance indicators including Total Sales Revenue, Average Property Price, Sold vs. Unsold Percentage, Unsold Inventory Value, Inventory Turnover Rate, Price per Square Foot, and Monthly Sales Trend using Excel formulas and pivot tables.
- To build an interactive Excel dashboard featuring KPI tiles, bar charts, pie charts, line graphs, and slicers to visualize sales performance by location, agent, property type, and time period.
- To integrate Microsoft Copilot to automate the generation of insights, summaries, and business recommendations from the analyzed data, documenting all prompts and AI-generated outputs.
- To answer the ten defined business questions covering revenue, agent rankings, location-wise performance, seasonal trends, and property type demand using data-driven analysis.
- To document the entire project workflow, upload all deliverables to a GitHub repository, and record a video demonstration of the working dashboard for final submission.

5. Methodology / Implementation Plan

The project follows a structured five-phase implementation plan to build the Real Estate Sales and Property Portfolio Dashboard. Each phase builds on the previous one, progressing from raw data exploration through to final dashboard delivery and submission.

#	Phase	Description

1	**Data Collection & Exploration**	The real estate dataset comprising 550 property records in Excel format was provided by Edunet Foundation. The dataset includes 16 fields: Property ID, Location, Property Type, Sale Price, Sale Date, Agent Name, Commission %, Property Status, Current Value, Occupancy Status, and property size among others. Initial exploration involved reviewing field names, data types, value distributions, and identifying data quality issues such as inconsistent formats and missing values.
2	**Data Cleaning & Preprocessing**	Data cleaning was performed in Excel to ensure accuracy and reliability. Price and date formats were standardized for consistent calculations. Duplicate property records were removed to prevent inflated sales totals. Missing values were addressed using logical assumptions and cross-referencing related columns. Categorical fields such as Property Type (e.g., standardizing 'Apartment', 'Apt', and 'Flat' into one category) and Occupancy Status were normalized to improve consistency across all visualizations.
3	**Analysis & AI Integration**	Excel pivot tables and formulas (SUMIF, COUNTIF, AVERAGEIF, VLOOKUP) were used to compute KPIs including Total Sales Revenue, Average Property Price, Location-wise Revenue, Sold vs. Unsold Percentage, Unsold Inventory Value, Price per Square Foot, and Monthly Sales Trend. Microsoft Copilot was used to generate automated business narratives, identify patterns, and produce strategic recommendations from the analyzed real estate data.
4	**Visualization / Output**	An interactive Excel dashboard was built featuring KPI summary tiles, bar charts for location-wise and agent-wise revenue, pie charts for property type distribution and sold vs. unsold breakdown, and a line graph for monthly sales trends. Slicers were added to allow filtering by location, property type, and time period. All Microsoft Copilot prompts and AI-generated insights were documented alongside the dashboard outputs.
5	**Documentation & Submission**	The complete project report was written covering data understanding, cleaning steps, KPI definitions, business insights, and Copilot usage. All project files including the cleaned dataset, Excel dashboard, and report were uploaded to a GitHub repository. A video demonstration of the working dashboard was recorded and submitted along with the GitHub and dashboard links for evaluation.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core i3 or higher, minimum 4 GB RAM (8 GB recommended)
- Operating System: Windows 10 / 11 or macOS
- Microsoft Excel (Microsoft 365 / Excel 2019 or later)
- Power BI Desktop — latest version (free download from Microsoft) — where applicable

- Microsoft Copilot — via Microsoft 365 subscription — where applicable
- AI Chatbot (ChatGPT / Google Gemini / Microsoft Copilot Web) — for prompt-based projects
- Internet connection — required for Copilot and AI Chatbot access
- Web browser — Google Chrome or Microsoft Edge (latest version)

B. Dataset

- The dataset is already available and was provided by Edunet Foundation as part of the AICW Fellowship Program. It is in Microsoft Excel format (.xlsx) containing 550 rows and 16 columns.
- Dataset fields include: Property ID, Location, Property Type (Residential / Commercial / Plot), Sale Price, Sale Date, Agent Name, Commission %, Property Status (Sold / Active), Current Value, Occupancy Status, and property area. This dataset enables analysis of sales trends, agent performance, and portfolio value across multiple locations.
- No additional data collection or simulation is required. The dataset is complete and has been validated for use in this project. All preprocessing will be performed directly in Microsoft Excel using built-in tools including Remove Duplicates, IF, IFERROR, Text-to-Columns, and pivot tables.
- Microsoft Copilot will be used as the AI tool to generate business insights, automated summaries, and strategic recommendations from the analyzed data. Copilot interactions will be documented with all prompts and corresponding AI outputs included in the project report.

C. Project Timeline

The following table outlines the planned tasks and expected completion timeline for the Real Estate Sales and Property Portfolio Dashboard project across a four-week period.

Phase	Task	Tentative Completion Date
1	Real estate dataset (550 records, 16 fields) collection, field review, and initial data quality assessment	*Week 1*
2	Data cleaning in Excel: removing duplicates, handling missing values, standardizing property types, dates, and price formats	*Week 2*
3	KPI calculations using pivot tables and Excel formulas; Microsoft Copilot integration for automated insights and business narrative generation	*Week 2 – 3*

4	Excel dashboard build with KPI tiles, charts, slicers for location, agent, and property type; testing and stakeholder review	*Week 3*
5	Report writing, GitHub upload, and video recording and submission	*Week 4*

7. Expected Outcomes / Conclusion

By the end of this project, the Real Estate Sales & Property Portfolio Dashboard will deliver a comprehensive, AI-assisted decision-support system tailored for the real estate industry. The outcomes are structured to directly address the five operational challenges identified:

- **Validated Dataset:** A fully cleaned and standardized dataset of 550 property records, with documented preprocessing steps ensuring accuracy, consistency, and reliability for analysis.
- **Centralized Dashboard:** An interactive Excel dashboard consolidating city-wise, agent-wise, and property-type performance into a unified view, enabling managers to detect underperforming markets and optimize strategies in real time.
- **KPI-Driven Insights:** Automated calculation and visualization of key performance indicators such as total revenue, unsold inventory value, agent productivity rankings, and seasonal demand cycles.
- **AI Integration:** Microsoft Copilot-generated narratives and recommendations embedded within the dashboard, providing actionable insights on pricing strategies, agent deployment, and inventory prioritization.
- **Business Impact:** A clear demonstration of how AI and data analytics can transform fragmented property records into a structured portfolio management tool, improving transparency, accountability, and profitability across multiple cities.
- **Final Deliverables:** A GitHub repository containing the cleaned dataset, dashboard file, and project report, along with a recorded video demonstration showcasing the dashboard's functionality and AI-driven insights.

This project not only equips stakeholders with a practical tool for monitoring and improving portfolio performance but also highlights the transformative role of AI in modern business intelligence. By bridging raw data with actionable insights, the dashboard ensures that real estate companies can make faster, evidence-based decisions and strategically align operations with market demand.

8. References

List all references, tutorials, and documentation used in the project. Students must cite any material they referred to, including research papers, GitHub repositories, YouTube tutorials, blogs, and official documentation.

- Edunet Foundation. AICW Fellowship Program Guidelines. 2026. Available at: https://www.edunetfoundation.org
- LinkedIn Learning. AI and Data Analytics Courses. 2026. Available at: https://www.linkedin.com/learning
- Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel
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